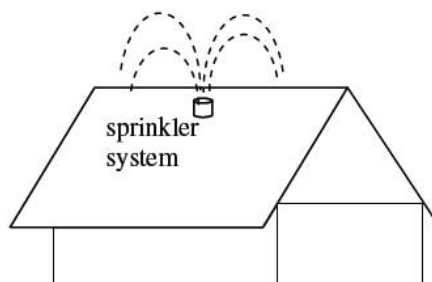


PP Q4

4. The sprinkler system on a rooftop is able to spray small water droplets onto the rooftop which can lower the temperature of the rooftop on hot sunny days. Which of the following explanations about the sprinkler system is/are reasonable ?



- (1) Water is a good conductor, which conducts heat quickly.
 - (2) Water has a high specific heat capacity, absorbing a lot of energy when its temperature rises.
 - (3) Water has a high specific latent heat of vaporization, absorbing a lot of energy when it evaporates.
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only

2013 Q2

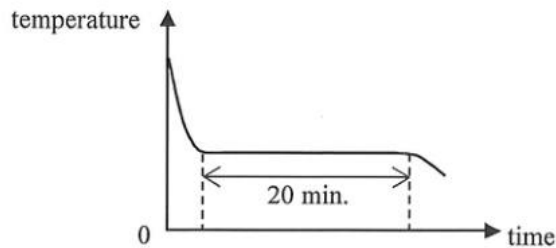
2. In an experiment to measure the specific latent heat of vaporization of water, a beaker of water is boiled off using an electric heater. Which of the following sources of error would lead to an experimental result smaller than the standard value ?
- A. Energy is lost to the surroundings.
 - B. Water splashes out of the beaker.
 - C. Steam condenses on the cooler part of the heater and drops back to the beaker.
 - D. The heater is not completely immersed in water.

2016 Q2

2. 0.3 kg of water at temperature 50°C is mixed with 0.2 kg of ice at temperature 0°C in an insulated container of negligible heat capacity. What is the final temperature of the mixture ?
- Given: specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$
 specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$
- A. -1.8°C
 - B. 0°C
 - C. 1.8°C
 - D. 3.0°C

2023 Q2

2.



A hot liquid substance of mass 0.3 kg and with heat capacity $600 \text{ J } ^\circ\text{C}^{-1}$ is left to cool in air. Its temperature falls at a rate of $2 \text{ }^\circ\text{C}$ per minute just before it begins to solidify. Then its temperature remains steady for 20 minutes until all the liquid just solidifies. Estimate the specific latent heat of fusion of the substance.

- A. 20000 J kg^{-1}
- B. 24000 J kg^{-1}
- C. 48000 J kg^{-1}
- D. 80000 J kg^{-1}

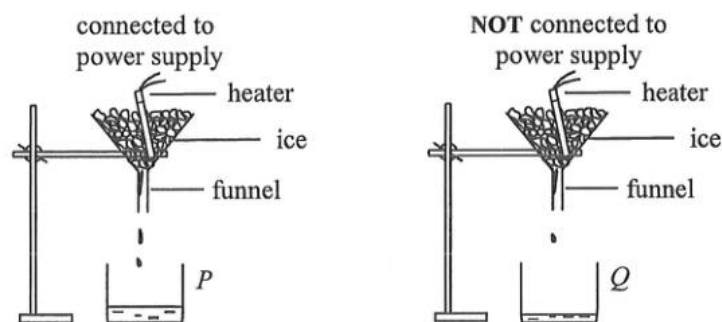
2024 Q2

2. If 0.5 kg of water at $50 \text{ }^\circ\text{C}$ is mixed with 2.0 kg of ice at $0 \text{ }^\circ\text{C}$, what is the final temperature of the mixture ?
 Given: specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$
 specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$

- A. $-54 \text{ }^\circ\text{C}$
- B. $0 \text{ }^\circ\text{C}$
- C. $10 \text{ }^\circ\text{C}$
- D. $25 \text{ }^\circ\text{C}$

2025 Q2

2. The set-up below can be used to find the specific latent heat of fusion of ice, l_f . The energy supplied by the heater is measured by a joulemeter. The mass of ice melted by the heater is given by the difference between the mass of water collected in beakers P and Q .



How would the calculated value of l_f be affected if

- (I) the heater is not completely covered by the ice ?
- (II) the mass of water collected in beaker P is taken as the mass of ice melted by the heater ?

- | | (I) | (II) |
|----|----------|----------|
| A. | decrease | increase |
| B. | decrease | decrease |
| C. | increase | increase |
| D. | increase | decrease |